Infix, Prefix & Postfix Expressions (BODMAS)

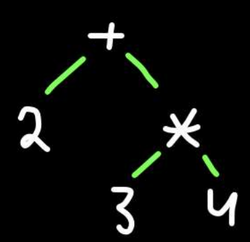
$x+y$ $\quad$ $+xy$ $\quad$ $xy+$

$\wedge \rightarrow$ priority 3

$/, * \rightarrow$ priority 2

$+, - \rightarrow$ priority 1

$2+3\times4$ $\quad$ $+2*34$ $\quad$ $234*+$



inorder : $2+3*4$

preorder : $+2*34$

postfix : $234*+$

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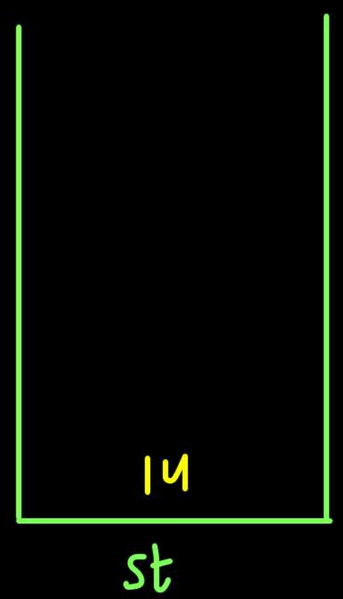
Postfix Evaluation using stack

s = "2 3 4 x +"




```
if( integer) st.push
if(operator){
    a = st.pop()
    b = st.pop()
    st. push (b operator a)
}
```

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Postfix to Infix (same process like postfix Evaluation)

2 3 4 x +

2 3 + 4 x

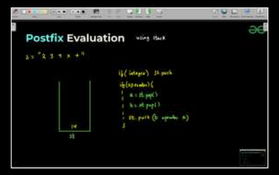
(Code - HW)

(2 + (3 x 4))

st<String>

((2 + 3) x 4)

st<String>



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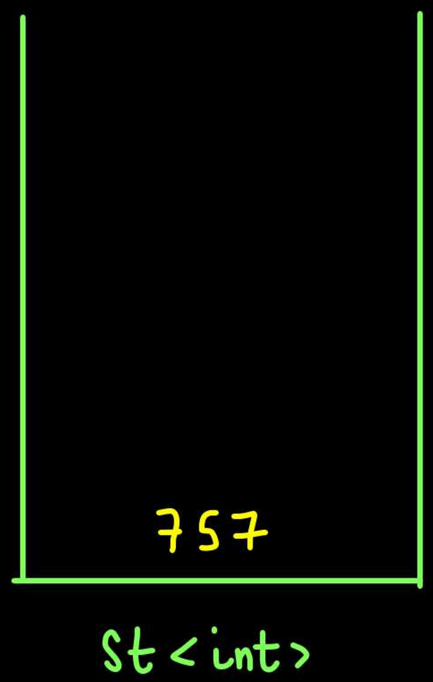
Prefix Evaluation

["+", "*", "/", "+", "100", "200", "2", "5", "7"]

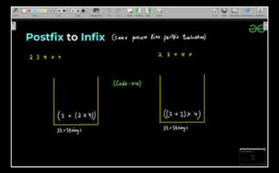


in: 2 - 3

pre: - 2 3

val = top op bottom

(Code - HW)



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Prefix to Infix

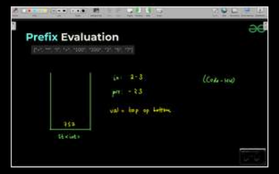
reverse

x + 2 3 4

((2 + 3) x 4)

st<string>

top op bottom



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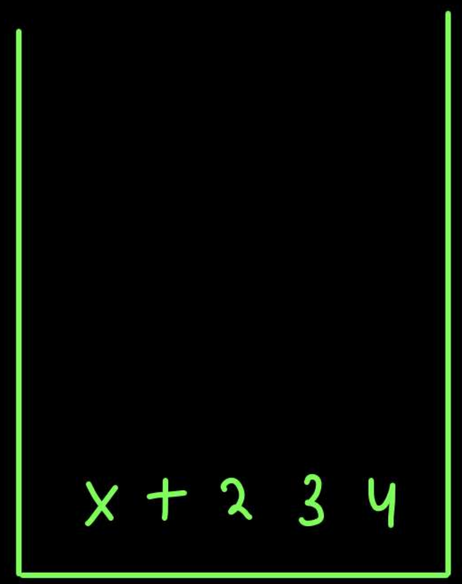
Postfix To Prefix

2 3 + 4 x

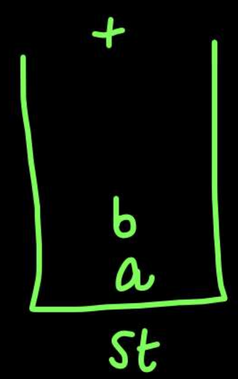
x + 2 3 4



st<string>

operator bottom top

a b +



st

operator bottom top

+ a b

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Prefix To Postfix

x + 2 3 4

(Code HW)

2 3 + 4 x

st<string>

top bottom operator

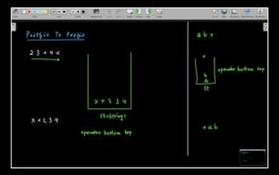
+ ab

+
a
b

st

top bt op

ab +



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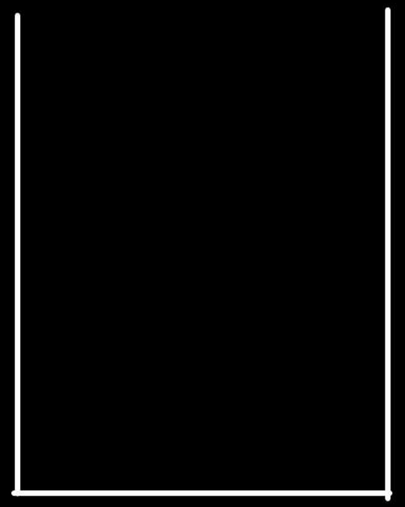
Infix to Postfix

सेल → priority

$2 \wedge 2 \wedge 3$ it means 2^{2^3}

$\wedge$	p-3
/, ×	p-2
+, -	p-1



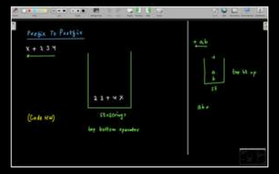
Stack
(operators)
(Exclusively)

ans = 2 2 3 ∧ ∧

op == ∧ ||

```
if ((priority(op) > priority(st.top)) st.push(op)
else ans += st.pop()
```

<=



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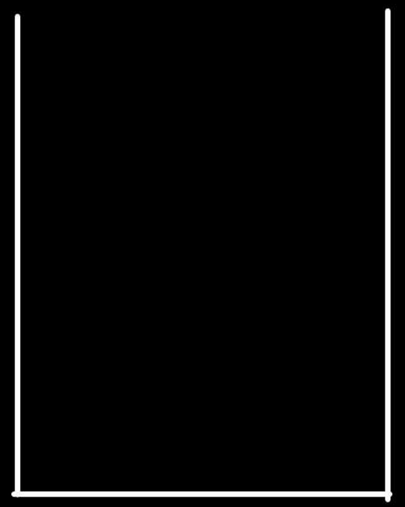
Infix to Postfix

सेल → priority

$(a+b) \times (c+d)$

^	p-3
/, ×	p-2
+, -	p-1
(	p-0



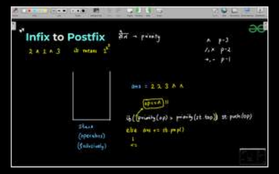
Stack
(operators)
(Exclusively)

ans = ab + cd + x

$\boxed{op == ^\wedge}$ ||

```
if((priority(op) > priority(st.top)) st.push(op)
else ans += st.pop()
```

<=



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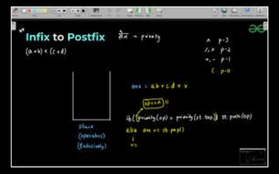
Infix to Prefix

HW Code Also



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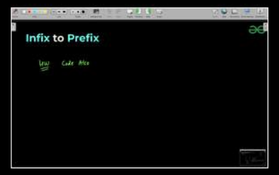
Infix Evaluation

HW

Code Also

9-11 AM Class.

Quick Sort



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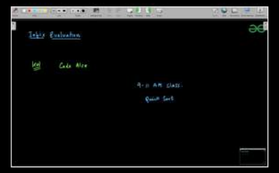
THANKYOU

Cuties